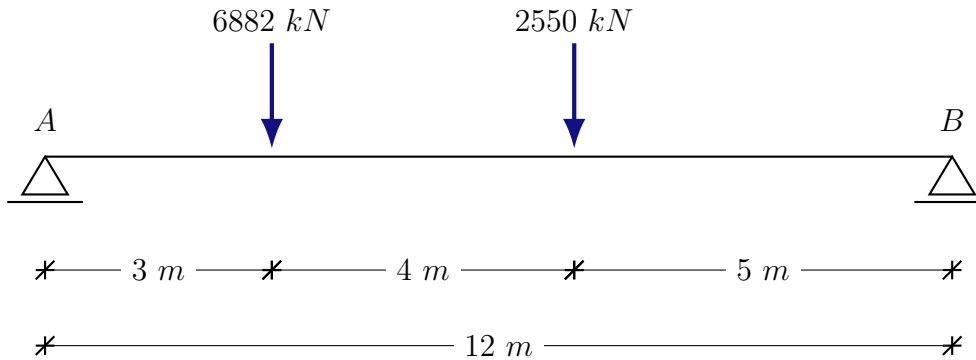


## Exercício

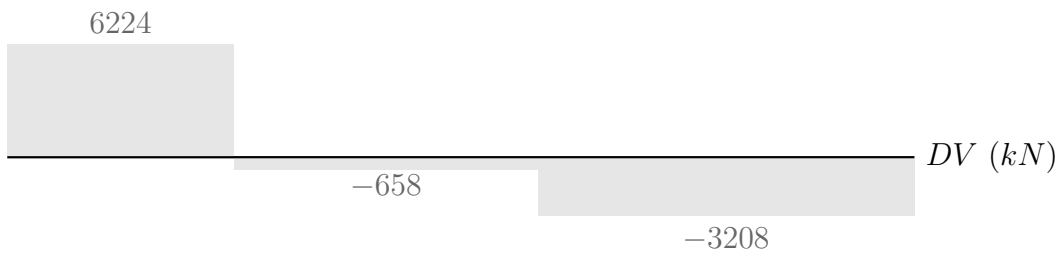
1. Dimensione a viga  $AB$  usando aço AR 350 COR, as cargas já estão majoradas escolha um perfil VS.



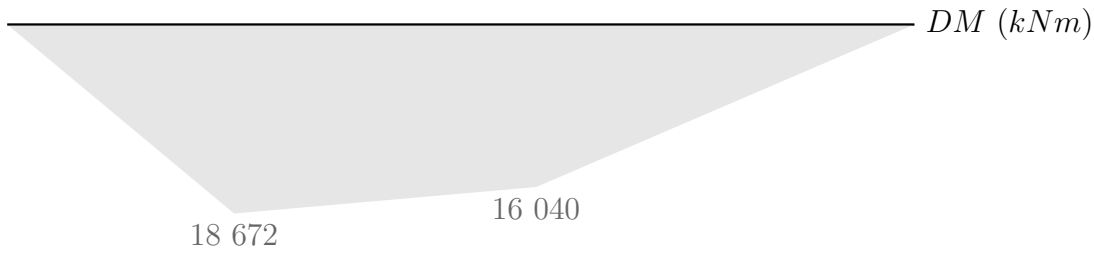
Reações de apoio e esforços solicitantes

$$\begin{aligned}\sum M_B &= 0 \\ -12V_A + 9 \cdot 6882 + 5 \cdot 2550 &= 0 \\ V_A &= 6224 \text{ kN}\end{aligned}$$

$$\begin{aligned}\sum F_y &= 0 \\ V_A + V_B - 6882 - 2550 &= 0 \\ V_B &= 3208 \text{ kN}\end{aligned}$$



$$V_{Sd,max} = V_A = 6224 \text{ kN}$$



$$M_{Sd,max} = M_C^1 = 18\,672 \text{ kNm}$$

### Propriedades do material e coeficiente

AR 350 COR  $\rightarrow f_y = 350 \text{ MPa}, f_u = 485 \text{ MPa}$  (Tabela A.1 da NBR 8800)

$\gamma_{a1} = 1,1$  (Tabela 3 da NBR 8800)

### Pré-dimensionamento à flexão

$\lambda \leq \lambda_p$  (Anexo D.2.1 da NBR 8800)

$M_{Sd} \leq M_{Rd}$  (Item 5.4.1.3 da NBR 8800)

$M_{Sd} \leq \frac{1}{\gamma_{a1}} M_{pl} \rightarrow M_{pl} = f_y W$  (Tabela D.1 da NBR 8800)

$$W = \frac{M_{Sd} \cdot \gamma_{a1}}{f_y}$$

$$W = \frac{18\,672 \cdot 10^3 \times 1,1}{350 \cdot 10^6}$$

$$W \approx 58,68 \cdot 10^{-3} \text{ m}^3$$

### Pré-dimensionamento ao cisalhamento

$\lambda \leq \lambda_p$  (Anexo D.2.1 da NBR 8800)

$V_{Sd} \leq V_{Rd}$  (Item 5.4.1.3 da NBR 8800)

$V_{Sd} \leq \frac{V_{pl}}{\gamma_{a1}} \rightarrow V_{pl} = 0,6 A_w f_y$  (Item 5.4.3.1.1, 5.4.3.1.2 da NBR 8800)

$$A_w = \frac{V_{Sd} \times \gamma_{a1}}{0,6 f_y}$$

$$A_w = \frac{6224 \cdot 10^3 \times 1,1}{0,6 \times 350 \cdot 10^6}$$

$$A_w = 32,60 \cdot 10^{-3} \text{ m}^2$$

### Perfil VS

$A_w = d \times t_w$  (Item 5.4.3.1.2 da NBR 8800)

$$A_w = 1500 \cdot 10^{-3} \times 12,5 \cdot 10^{-3}$$

$$A_w = 18,75 \cdot 10^{-3} \text{ m}^2$$

$$VS\ 1500 \times 492 \rightarrow W = 38950 \text{ cm}^3, A = 18,75 \cdot 10^{-3} \text{ m}^2$$

∴ Não existe perfil VS capaz de suportar esse carregamento. Seria necessário projetar uma sob medida.

## CORREÇÃO

### Pré dimensionamento à flexão

$$\begin{aligned} M_{Sd} &\leq M_{Rd} \\ 18672 \cdot 10^3 &\leq \frac{W \times 350 \cdot 10^6}{1,1} \\ W &\approx 58,68 \cdot 10^{-3} \text{ m}^3 \end{aligned}$$

Perfil escolhido: VS  $800 \times 143 \rightarrow Z = 5910 \text{ cm}^3$

### FLM

$$\begin{aligned} \lambda &\leq \lambda_p \\ \frac{b}{t} &\leq 0,38 \sqrt{\frac{E}{f_y}} \\ \frac{\frac{320 \cdot 10^{-3}}{2}}{19 \cdot 10^{-3}} &\leq 0,38 \sqrt{\frac{200\,000 \cdot 10^6}{350 \cdot 10^6}} \\ 8,42 &\leq 9,08 \end{aligned}$$

$$M_{Rd} = \frac{M_{pl}}{\gamma_{a1}} = \frac{5910 \times 35}{1,1} = 188045,45 \text{ kNcm}$$

### FLA

$$\lambda = \frac{h}{t_w} = \frac{76,2}{0,8} = 95,25$$

$$\lambda_p = 3,76 \sqrt{\frac{20000}{35}} = 89,88$$

$$\lambda_r = 5,7 \sqrt{\frac{20000}{35}} = 136,25$$

$$\lambda_p \leq \lambda < \lambda_r$$

$$M_r = w_x \times f_y = 5374 \times 35 = 188090 \text{ kNcm}$$

$$M_{pl} = 206850 \text{ kNcm}$$

$$M_{Rd} = \frac{1}{\gamma_{a1}} = (M_{pl} - (M_{pl} - M_r) \frac{\lambda - \lambda_p}{\lambda_r - \lambda_p}) = 186070,53 \text{ kNcm}$$

$\therefore$  Acontecerá flambagem na alma desse perfil.

## SEGUNDA TENTATIVA

### FLM

$0,35 \leq k_c \leq 0,76$  (Tabela 4, nota de rodapé a da Norma 8800)

$$k_c = \frac{4}{\sqrt{\frac{h}{t_w}}}$$

$$k_c = \frac{4}{\sqrt{\frac{81,2}{0,8}}}$$

$$k_c = 0,397$$

$$\lambda_p < \lambda \leq \lambda_r$$

$$0,38 \sqrt{\frac{E}{f_y}} < \frac{b}{t} \leq 0,95 \sqrt{\frac{E}{(f_y - \sigma_r)/k_c}}$$

$$0,38 \sqrt{\frac{200\,000 \cdot 10^6}{350 \cdot 10^6}} < \frac{\frac{350 \cdot 10^{-3}}{2}}{19 \cdot 10^{-3}} \leq 0,95 \sqrt{\frac{200\,000 \cdot 10^6}{(350 \cdot 10^6 - \sigma_r)/0,397}}$$

$$9,08 < 9,21 \leq 17,10$$

$$M_{pl} = 6845 \times 35 = 2396 \text{ kNcm}$$

$$M_r = 6243 \times 35 = 218505 \text{ kNcm}$$

$$M_{Rd} = \frac{1}{\gamma_{a1}} = (M_{pl} - (M_{pl} - M_r) \frac{9,21 - 9,08}{17,12 - 9,08}) = 217485 \text{ kNcm}$$

### FLA

$$\lambda = \frac{b}{t_w} = \frac{81,2}{0,8} = 101,5$$

$$\lambda_p = 89,88$$

$$\lambda_r = 136,25$$

$$M_{Rd} = \frac{1}{\gamma_{a1}} = (M_{pl} - (M_{pl} - M_r) \frac{9,21 - 9,08}{17,12 - 9,08}) = 212995 \text{ kNcm}$$

### FLT

$$\lambda = \frac{L_b}{r_y} = \frac{1200 \cdot 10^{-2}}{8,28 \cdot 10^{-2}} = 144,92$$

$$\lambda = 1,76 \sqrt{\frac{E}{f_y}} = 42,07$$

$$\lambda_r = \frac{1,38 c_b \sqrt{I_y J}}{r_y \times J \times B_y} \sqrt{1 + \sqrt{\frac{1 + 27 C_w B_1^2}{C_b^2}}} = 118$$

$$B_1 = \frac{(f_y - \sigma_r)W}{EJ}$$

$$B_1 = \frac{(\frac{70}{100}f_y)6243}{20\,000 \times 174}$$

$$B_1 = 0,0439$$

$$C_w = \frac{I_y(d - t_y)^2}{4} \text{ Para seções I}$$

$$C_w = \frac{13581(85 - 1,9)^2}{4}$$

$$C_w = 2344672,35$$

$$C_b d = \frac{12,5M_{max}}{2,5M_{max} + 3M_A + 4M_B + 3M_C} R_m \rightarrow R_M = 1$$

$$C_b = \frac{12,5 \times 186720}{2,5 \times 186720 + 3 \times 186720 + 4 \times 16698 + 3 \times 9624} R_m$$

$$C_b = 1,176$$

$$M_{Rd} = \frac{1}{\gamma_{a1}} M_{Ce}$$

## TERCEIRA TENTATIVA

Perfil VS 900 x 177

$$\lambda = \frac{h}{t_w} = \frac{85,5 \text{ cm}}{0,8 \text{ cm}} = 106,875$$

$$\lambda_p = 1,1 \sqrt{\frac{K_v \cdot E}{f_y}} \rightarrow K_v = 5,34$$

$$\lambda_p = 60,76$$

$$\lambda_r = 1,37 \sqrt{\frac{k_v \cdot E}{f_y}} = 75,67$$

$$\lambda > \lambda_r$$

$$V_{Rd} = 1,24 \left( \frac{\lambda_p}{\lambda} \right)^2 \frac{V_{pl}}{\gamma_{a1}} = 1,24 \left( \frac{60,76}{106,875} \right)^2 \frac{1511,66}{1,1} = 550,76 \text{ kN}$$

$$V_{pl} = 0,6 A_w f_y = 1511,66 \text{ kN}$$

$$A_w = d \cdot t_w = (h_0 2 t_f) t_w = 71,984 \text{ cm}$$

$$V_{Sd} > V_{Rd} \rightarrow \text{n\~ao suporta.}$$

$$\frac{a}{h} > 3$$

$$a = 3,1 \cdot h = 2,9 \times 85,5$$

$$265 \text{ cm}$$

$$k_v = 5 + \frac{5}{(a/h)^2} = 5 + \frac{5}{(265/85,5)^2} = 5,52$$

$$k_v = 5 + \frac{5}{(a/h)^2} = 5 + \frac{5}{(100/85,5)^2} = 8,65$$

$$\lambda_p = 77,33$$

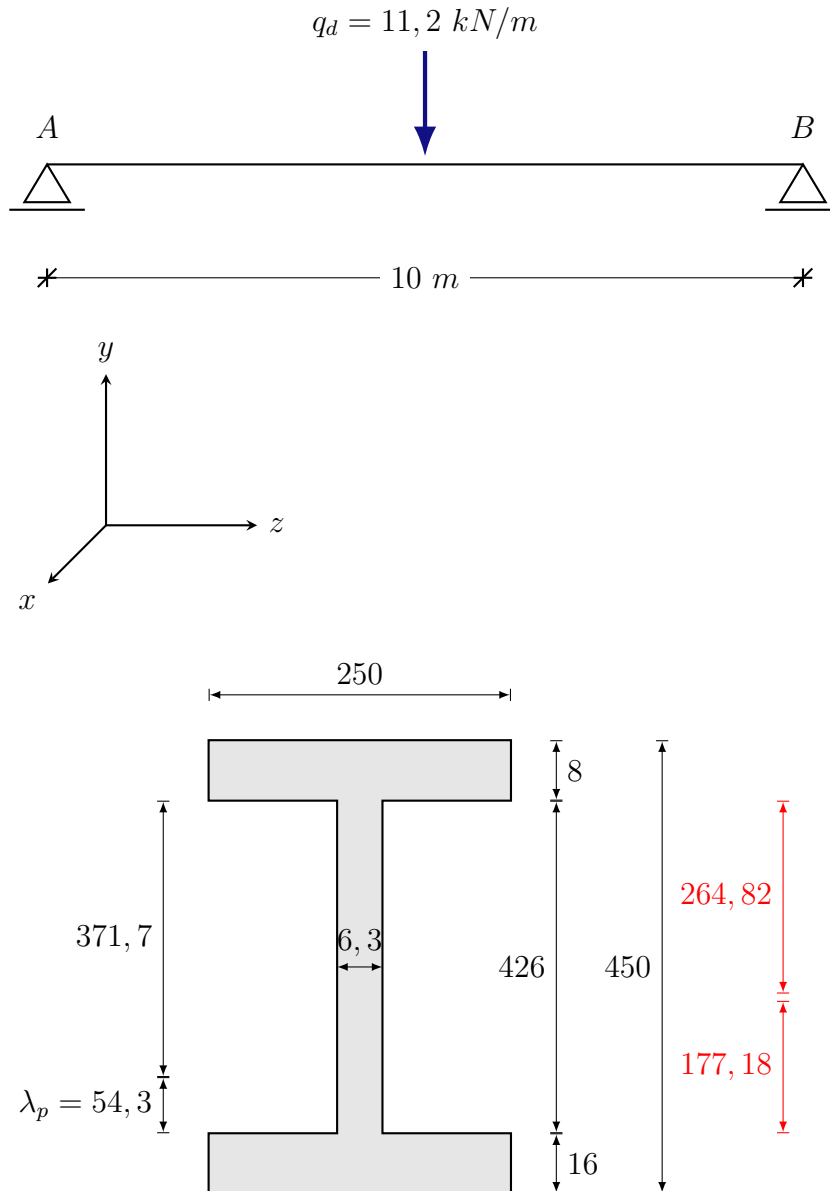
$$\lambda_r = 96,31$$

$$V_{Rd} = 892,127 \text{ kN}$$

$$0,56 \sqrt{\frac{E}{f_y}}$$

## Exercícios

1. Verificar  $FLA$ ,  $FLM$ ,  $FLT$  para a viga birrotulada VSM 450 x 68, aço USICIVIL 350. Considere já os valores já majorados.



Dimensões em  $mm$ .

$$A_g = 86,8\text{ cm}^2$$

$$I_y = 31,26\text{ cm}^4$$



$$\begin{aligned}
r_y &= 6 \text{ cm} \\
I_x &= 30691 \text{ cm}^4 \\
C_w &= 1332250 \text{ cm}^6 \\
W_{xs} &= W_{xc} = 1125 \text{ cm}^3 \\
W_{xi} &= W_{xc} = 1732 \text{ cm}^3 \\
Z &= 1445 \text{ cm}^3 \\
J &= 42 \text{ cm}^4
\end{aligned}$$

FLA - tabela D1 p. 144

$$\lambda = \frac{h_c}{t_w} = \frac{2 \times 264,82}{6,3} = 84,06$$

$$\lambda_p = \frac{\frac{h_c}{h_p} \sqrt{\frac{E}{f_y}}}{(0,54 \frac{M_{pl}}{M_r} - 0,09)^2} \leq \lambda_r$$

$$\lambda_p = \frac{\frac{52,96}{74,34} \sqrt{\frac{20000}{35}}}{(0,54 \frac{50575}{39375} - 0,09)^2} \leq \lambda_r$$

$$\lambda_r = 46,74 \leq \lambda_r$$

$$\lambda_r = 5,70 \sqrt{\frac{E}{f_y}} = 5,70 \sqrt{\frac{20000}{35}} = 136,25$$

$$y_{CG} = \frac{\sum Ay}{\sum A}$$

$$A_1 = 8 \times 250 = 2000 \text{ mm}^2$$

$$A_2 = 6,3 \times 426 = 2683,8 \text{ mm}^2$$

$$A_3 = 16 \times 250 = 4000 \text{ mm}^2$$

$$y_1 = 16 + 426 + 8/2 = 445 \text{ mm}$$

$$y_2 = 16 + 426/2 = 229 \text{ mm}$$

$$y_3 = 16/2 = 8$$

$$y_{CG} = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3}{A_1 + A_2 + A_3} = 177,179 \text{ mm}$$

$$h_p = 2 \times 37,17 = 74,34 \text{ cm}$$

$$M_{pl} = Z \times f_y = 1445 \times 35 = 50575 \text{ kNcm}$$

$$M_r = W \times f_y = 1125 \times 35 = 39375 \text{ kNcm}$$

Compacto  $\lambda$

Semi  $\lambda_p$

Esbelto  $\lambda_r$

D 2.2

$$M_{Rd} = \frac{1}{\gamma_{a1}} \left( M_{pl} - (M_{pl} - M_r) \frac{\lambda - \lambda_p}{\lambda_r - \lambda_p} \right) = 41732,09 \text{ kNcm}$$

$$M_{Sd} = \frac{0,112 \times 1000^2}{8} = 14000 \text{ kNcm}$$

FLA OK

$$M_{Sd} \leq M_{Rd} \rightarrow 14000 \leq 41732$$

FLM - tabela D.1 p. 144

$$\lambda = \frac{b}{t} = \frac{\frac{250}{2}}{8} = 15,62 \text{ ver D.2.8-h}$$

$$\lambda_p = 0,38 \sqrt{\frac{E}{f_y}} = 0,38 \sqrt{\frac{20000}{35}} = 9,08$$

D2.8f - perfis soldados

$$\lambda_r = 0,95 \sqrt{\frac{E \times K_c}{70\% f_y}} = 0,95 \sqrt{\frac{20000 \times 0,48}{0,7 \times 35}} = 18,8$$

$$k_c = \frac{4}{\sqrt{\frac{h}{t_w}}} = \frac{4}{\sqrt{\frac{4,26}{0,63}}} = 0,48$$

$$\lambda_p \leq \lambda < \lambda_r$$

$$M_{Rd} = \frac{1}{1,1} \left( 50575 - (50575 - 39375) \frac{15,62 - 9,08}{18,8 - 9,08} \right) = 39126,54 \text{ kNcm}$$

FLM OK

$$M_{Sd} \leq M_{Rd}$$

FLT tabela D.1

$$\lambda = \frac{L_b}{r_{ytc}} = \frac{1000}{4,9} = 204,08$$

$$\lambda_p = 1,76 \sqrt{\frac{E}{f_y}} = 42,07$$

D.2.8b

$$\lambda_r = \frac{1,38C_b\sqrt{I_yJ}}{r_{ytc}JB_1} \sqrt{B_2 + \sqrt{B_2^2 + \frac{27C_wB_1^2}{C_b^2I_y}}}$$

$$I_{Tcy} = \frac{(0,8 \times 25^3) + (37,17 \times 0,63^3)}{12} = 1042,44 \text{ cm}^4$$

$$A_{gt} = (8 \times 25) + (63 \times 37,17) = 43,4171$$

$$r_{ytc} = \sqrt{\frac{I_{Tcy}}{A_g}} = 4,9 \text{ cm}$$

$$C_b = \frac{12,5 M_{mx}}{2,5M_{mx} + 3M_A + 4M_B + 3M_C} R_m$$

$$M_{max} = M_B = \frac{ql^2}{8} = \frac{11,2 \text{ kN/m} \times 10^2}{8} = 14000 \text{ kNcm}$$

$$M_A = M_C = \frac{3ql^2}{32} = \frac{qlx}{2} - \frac{qx^2}{2} = \frac{3 \times 0,112 \times 1000^2}{32} = 10500 \text{ kN}$$

$$R_m = 0,5 + 2\left(\frac{I_{ym}}{I_y}\right)^2 = 0,5 + 2\left(\frac{\frac{8,25^3}{12}}{3126}\right)^2 = 0,72$$

$$C_b = 0,82$$

$$B_1 = \frac{(0,7f_y)w_c}{EJ} \leq \frac{f_yw_c}{EJ} \rightarrow B_1 = 0,0328 \leq 0,0072$$

$$B_3 = 0,45\left(d - \frac{t_{fs} + t_{fi}}{2}\right)(2\alpha_y - 1) = 0,45\left(45 - \frac{0,8 + 1,6}{2}\right)(2 \times 0,33 - 1) = -6,70$$

$$\alpha_y = \frac{I_{ytc}}{I_y} = 0,33$$

$$\beta_2 = \frac{5,2\beta_1\beta_3}{C_b} + 1 = -0,3938$$

$$\lambda_r = 119,63$$

$$M_{cr} = 11833,82$$

$$M_{Rd} = \frac{1}{\gamma_{a1}} M_{cr} = 10758 \text{ kNcm}$$

2. Verificar a viga em balanço, mesma viga do exercício anterior.

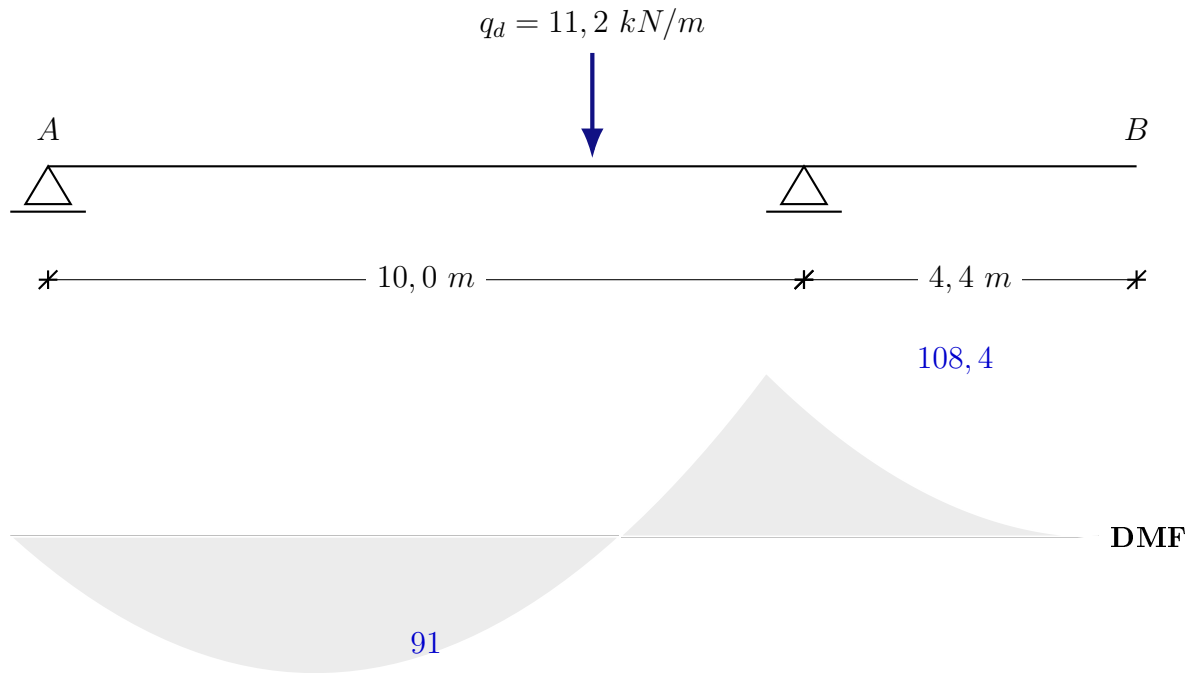
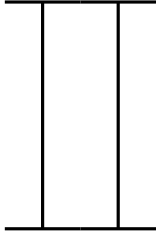


Diagrama de Momento Fletor.

Verificar FLA, FLM, e FLT.

3. Qual é o máximo momento fletor e cortante que uma viga de 20 m suporta? Com perfis justapostos. Carga? Considere carga pontual; 2 (VS 500 x 61).



FLM

$$\lambda \leq \lambda_p$$

$$M_{Rd} = \frac{M_{pl}}{\gamma_1}$$

$$\lambda = \frac{b}{t}$$

$$\lambda = \frac{\frac{b}{2}}{t}$$

FLA

$$\lambda_p \leq \lambda \leq \lambda_r$$



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## Trabalho

Passo a passo do cálculo de ligações parafusadas e soldadas. Valor (1,5 pontos).

Exercício escolher um entre os itens abaixo:



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## Exercícios

Perfil caixão Duplo perfil I

Exercícios propostos

FLEXÃO CORTE

6.2.4 7.2.1

6.2.5 7.3.1

6.3.1 7.3.3

6.3.4

Três questões da acima, e são três questões na prova